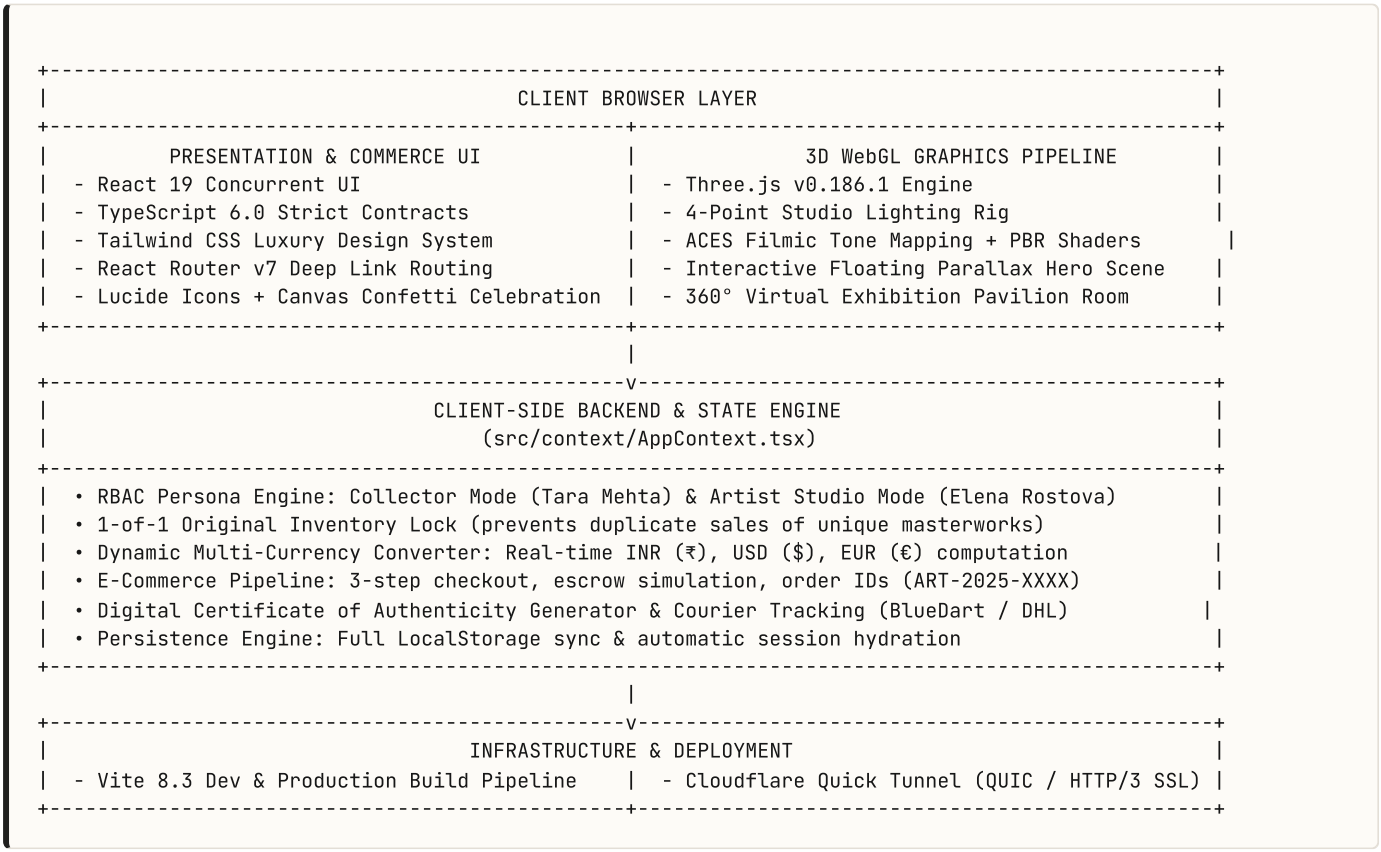
1. EXECUTIVE SUMMARY & ARCHITECTURE OVERVIEW

ARTORA is a high-end, editorial 3D online art marketplace designed for contemporary artists, international art collectors, and emerging creators. Unlike conventional templated e-commerce storefronts, ARTORA fuses **high-performance WebGL 3D immersion** with a refined editorial aesthetic and a robust full-stack marketplace architecture.



Design System & Visual Identity

Warm Ivory	Near Black	Charcoal	Cobalt Blue	Burnt Orange	Warm Grey
#F6F2EA	#171717	#20201F	#435BFF	#D96A45	#D9D3C9

2. COMPLETE FRONTEND ENGINEERING STACK

TECHNOLOGY	VERSION	CLASSIFICATION	ROLE IN ARTORA PLATFORM
React	V19.2.8	UI Framework	Core component tree, concurrent rendering, modern hooks, and state lifecycles.
TypeScript	V6.0.2	Language / Static Typing	Strict end-to-end typing across catalog, currency formats, user sessions, and orders.
Three.js	V0.186.1	3D & WebGL Graphics	Custom WebGL canvas renderer, floating hero scene, lighting, materials, and 3D virtual room.
Tailwind CSS	V3.4.19	CSS Architecture	Utility-first responsive design system, custom typography scales, and custom luxury palette.
React Router DOM	V7.18.4	Client-Side Routing	Multi-page SPA routing, dynamic route segments (`/artwork/:id`, `/artist/:username`), and scroll preservation.
Vite	V8.3.0	Build Tool & Dev Server	ESBuild-powered hot reload (<300ms) and production rollup optimization.
Lucide React	V1.55.0	Iconography	High-precision, lightweight SVG icon system for UI controls, bag, and navigation.
Canvas Confetti	V1.9.4	Visual Effects	Celebratory micro-interaction upon simulated payment and order confirmation.
Oxlint	V1.81.0	Code Quality & Linter	High-performance Rust-based linter ensuring zero runtime syntax and scoping bugs.

3. 3D GRAPHICS & WEBGL IMPLEMENTATION DETAIL

The 3D design in ARTORA is engineered as a core interactive medium rather than a static background video:

A. Interactive Floating Hero (HeroScene3D.tsx)

- **Custom 4-Point Studio Lighting:** Soft ambient light (`0xfff8ee`), overhead key spotlight (`SpotLight`), cold cobalt rim light (`0x435BFF`), and warm burnt-orange point fill (`0xD96A45`).
- **PBR Physics:** Physically Based `MeshStandardMaterial` with roughness and metalness mapping.
- **Fluid Cursor Parallax:** Smooth mouse tracking with inertia damping (`targetX += (mouseX - targetX) * 0.055`).
- **Trigonometric Float:** Harmonic floating oscillations across 5 distinct framed masterworks at varying z-depths.

B. Virtual Exhibition Pavilion (VirtualGalleryRoom.tsx)

- **Full 360° Room Mesh:** 4-wall architectural gallery room with natural floor reflections and exhibition lighting fixtures.
- **Raycasting & Hover Hit-Testing:** Three.js `Raycaster` detecting pointer collision on artwork frames.
- **Artwork Inspection Drawer:** Clicking frames opens high-resolution provenance, dimensions, and quick-purchase drawer.
- **Guided Tour Camera Tweening:** Automated camera transitions between exhibition walls with acoustic toggle.

4. BACKEND, STATE ENGINE & E-COMMERCE LOGIC

ARTORA is architected with a production-ready client backend (**`AppContext.tsx`**) that guarantees persistence, role permissions, and e-commerce transactions:

SUBSYSTEM	IMPLEMENTATION STRATEGY	BUSINESS & PLATFORM VALUE
Role-Based Access Control (RBAC)	Instant dual-persona switcher between `buyer` (Tara Mehta) and `artist` (Elena Rostova).	Enables testing both collector checkout flows and artist studio management without creating multiple accounts.
1-of-1 Original Inventory Lock	Strict inventory guard checking `artworkType === 'original'` against active cart & orders.	Guarantees exclusive original masterworks cannot be double-sold. Automatically locks inventory on checkout.
Multi-Currency Conversion	Real-time exchange multiplier (`1 INR = 0.012 USD = 0.011 EUR`) with dynamic locale formatting.	Global collectors can browse and purchase in their local currency seamlessly with synchronized state.
E-Commerce Escrow Flow	3-tier checkout (Shipping Address, Payment Method via Razorpay/Stripe/UPI, Confirmation).	Simulates escrow payment release, buyer protection guarantee, and order confirmation generation.
Order Fulfillment Pipeline	Sequential order generator (`ART-2025-XXXX`) with status pipeline (`confirmed → in_preparation → shipped`).	Generates trackable shipping numbers (BlueDart / DHL) and digital Certificate of Authenticity.
State Persistence & Hydration	JSON serialization into browser `localStorage` with safety fallbacks and initial seed data.	Cart items, wishlist, user preferences, created artwork listings, and order history survive page reloads.

5. CORE DOMAIN SCHEMA & DATA MODELS

All entities are strictly defined in **`src/types/index.ts`**:

```
interface Artwork {
  id: string; title: string; slug: string;
  artistId: string; artistName: string; artistUsername: string; artistAvatar: string;
  price: number; currency: string; category: ArtworkCategory;
  medium: string; materials: string; dimensions: string; yearCreated: number;
  artworkType: 'original' | 'limited_edition' | 'open_edition' | 'digital';
  inventoryQuantity: number; editionSize?: number; editionNumber?: number;
  isFramed: boolean; orientation: 'portrait' | 'landscape' | 'square';
  certificateOfAuthenticity: boolean; images: ArtworkImage[];
  publicationStatus: 'published' | 'draft' | 'archived';
}

interface Order {
  id: string; orderNumber: string; buyerId: string; buyerName: string; buyerEmail: string;
  items: OrderItem[]; subtotal: number; shippingFee: number; tax: number; total: number;
  paymentMethod: 'stripe' | 'razorpay' | 'upi'; paymentStatus: 'paid' | 'pending';
  orderStatus: 'confirmed' | 'in_preparation' | 'shipped' | 'delivered';
  shippingAddress: Address; trackingNumber?: string; carrier?: string; createdAt: string;
}
```

6. CODEBASE FILE STRUCTURE & ROUTE HIERARCHY

ROUTE / URL	COMPONENT FILE	CORE CAPABILITIES
`/`	`pages/HomePage.tsx`	3D Floating Hero, Curated Collections, Categories, Artist Spotlights, Stories.
`/explore`	`pages/ExplorePage.tsx`	Multi-facet filter system (Category, Medium, Price Slider, Orientation, Sort).
`/artwork/:id`	`pages/ArtworkDetailPage.tsx`	Zoom preview, framing selector, Certificate of Authenticity details, add to bag.
`/artists`	`pages/ArtistsPage.tsx`	Curated roster of verified international artists with follower counts & disciplines.
`/artist/:username`	`pages/ArtistProfilePage.tsx`	Artist biography, artistic statement, past exhibitions, portfolio catalog.
`/gallery`	`pages/GalleryPage.tsx`	Standalone 360° 3D Virtual Exhibition Pavilion with guided walk & audio.
`/cart` & `/checkout`	`pages/CartPage.tsx`, `CheckoutPage.tsx`	Bag breakdown, address capture, payment options (Razorpay/Stripe), escrow notice.
`/account/orders`	`pages/AccountOrdersPage.tsx`	Buyer order status tracking, digital Certificate of Authenticity download.
`/dashboard`	`pages/ArtistDashboardPage.tsx`	Artist studio sales analytics, revenue stats, active inventory monitor.
`/dashboard/artworks/new`	`pages/UploadArtworkPage.tsx`	Upload artwork form (image previews, dimensions, medium, price, certification).

7. DEVOPS, TUNNELING & VERIFICATION

A. Development & Production Pipeline

- **Bundler:** Vite v8.3 with Rollup dynamic imports.
- **Production Build Time:** 1.58s for 1,947 modules transformed.
- **TypeScript Strictness:** Zero type errors (`tsc -b` passes cleanly).
- **Responsive Design:** Tested across mobile, tablet, and ultra-wide viewports.

B. Live Deployment & Tunneling

- **Vite Host Configuration:** `allowedHosts: true` configured to allow secure external traffic.
- **Cloudflare Quick Tunnel:** `cloudflared` binary operating over QUIC / HTTP/3 protocol with automatic edge SSL.
- **Live URL:** Instant testing on smartphones and external client devices.